

Multiple Choice (10 marks)

1. Which of the following is an incorrect statement about gel electrophoresis?
 - (a) DNA migrates from positive charge to negative charge.
 - (b) Smaller DNA travels faster.
 - (c) The DNA migrates only when the current is running.
 - (d) The longer the current is running, the farther the DNA will travel.
2. During the period when life is believed to have begun, the atmosphere on primitive Earth contained abundant amounts of all the following gases EXCEPT
 - (a) hydrogen
 - (b) methane
 - (c) nitrogen
 - (d) sulfur dioxide
 - (e) neon
3. Mutation of homeotic cluster genes often results in which of the following developmental defects in *Drosophila*?
 - (a) Complete absence of segmentation
 - (b) Hindrance in the development of the nervous system
 - (c) Inability to undergo metamorphosis
 - (d) Absence of a group of contiguous segments
 - (e) Tumor formation in imaginal discs
4. Which of the following would most likely provide examples of mitotic cell divisions?
 - (a) cross section of muscle tissue
 - (b) longitudinal section of a shoot tip
 - (c) longitudinal section of a leaf vein
 - (d) cross section of a fruit
 - (e) cross section of a leaf
5. Vinegar is classified as an acid. What does this tell you about the concentration of the hydrogen ions and the hydroxide ions?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions

- (c) Vinegar has a higher concentration of hydroxide ions than water
- (d) Vinegar has no hydroxide ions
- (e) Vinegar does not contain hydrogen ions

True / False (10 marks)

Answer True or False

- 6. True or False: Hemoglobin's affinity for O_2 decreases as blood pH decreases due to increased H^+ concentration.
- 7. True or False: The high pH levels in the stomach prevent its own secretions from causing digestion of the stomach lining.
- 8. True or False: Only the size of the wings influences a bird's flight characteristics, not their shape.
- 9. True or False: Spindle fibers are composed of microtubules, which are essential for the proper segregation of chromosomes during mitosis.
- 10. True or False: Silk worms spin cocoons based on the phase of the moon, demonstrating the influence of internal body secretions on their behavior.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the concept of stabilizing selection in the context of human birth weight, explaining how it influences the average weight around 3 kg and its implications for evolutionary fitness.
- 12. Discuss the concept of a "society" in biology, using a flock of birds as an example. Analyze how their communication and cooperative behaviors contribute to their survival.

End of Paper